

Scenario

A patient and a robot collaborate to produce a painting together, with the shared goal of creating an artwork that reflects the patient's mood. The robot contributes either by suggesting or applying specific colors/shapes, while the patient directs the creative process. Neither can achieve the final product alone, the robot provides structure/variation, the patient provides meaning/emotion.

Research

- **Research question:** Does a proactive robot collaborator (suggests colors/shapes unprompted) improve patient engagement compared to a reactive collaborator (only acts after the patient directs it)?
- **Hypothesis:** Proactive behavior increases perceived collaboration quality and reduces moments of stagnation.
- **Method:** Use Wizard-of-Oz, maybe in an online canvas where the participant can draw and collaborate with the "robot", the other user that is drawing. Run participants through two painting sessions (proactive vs reactive). Collect self-reports (engagement, enjoyment, trust) and maybe task-level metrics (number of contributions, smoothness of turn-taking, quality of the artwork?). So a within-subject design where the order of the conditions is randomized.
- **Measures:**
 - **Subjective:** Self-reports (engagement, enjoyment, trust, happiness with result) on a likert scale (based on an existing method), analysis by using one-way ANOVA to see whether there's a significant difference between the conditions
 - **Objective:** Time spent drawing total/pen on paper, idle time, analysis by using one-way ANOVA to see whether there's a significant difference between the conditions

Task Description

The participant is asked to draw something that reflects their current emotions.

Robot states

- **Proactive:** e.g. start painting initially, potentially recognizable shapes to associate with
- **Reactive:** user/patient starts painting, robot reacts and continues what they start

Possible sources

Peter Schaldenbrand, Gaurav Parmar, Jun-Yan Zhu, James McCann, & Jean Oh. "CoFRIDA: Self-Supervised Fine-Tuning for Human-Robot Co-Painting." 2024 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2024

Cooney, M., & Berck, P. (2019, May). Designing a robot which paints with a human: visual metaphors to convey contingency and artistry. In *ICRA-X Robots Art Program at IEEE International Conference on Robotics and Automation (ICRA)* (p. 2).